Energy, Work & Power

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A force of 42 N acts at 30° to the direction of motion of a body that moves 27 m. Find the work done, leaving your answer in exact (surd) form.

Final Answer: $1134 \times \frac{\sqrt{3}}{2}$ J. $W = Fd\cos 30^\circ = 42 \times 27 \times \frac{\sqrt{3}}{2} = 1134 \times \frac{\sqrt{3}}{2}$ J

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A body of mass 11 kg moving at 7 m/s has work 356 J done on it by a driving force (no resistance). Find its final speed, using the work-energy principle.

Final Answer: 10.66. $12mv^2 = 12mu^2 + Wv^2 = u^2 + 2Wm = (7)^2 + 2(356)11 = 1251/11v10.66$ m/s

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Normal Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Permutations & Combinations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Circular Measure · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Statistics & Probability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In how many ways can 5 people be seated in a row if two particular people (A and B) must not sit next to each other?

Total $= 5! = 120$. Cases where A & B are adjacent: treat AB as a block $\rightarrow 4! \times 2 = 48$. Answer: $120 - 48 = 72$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A data set has mean 12 and standard deviation 4. A new value of 20 is added to $n = 9$ existing values. Without finding the new variance, find the new mean.

Old sum $= 9 \times 12 = 108$. New sum $= 108 + 20 = 128$. New mean $= 128/10 = 12.8$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$.
$\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$.
Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}}$,.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Practice 4.2: Bivariate data for study hours x and exam mark y across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of y on x has equation $y = 2.4x + c$. (a) Find the value of c. (b) Interpret the slope 2.4 in context. (c) Estimate the exam mark of a student who studies 18 hours.

(a) The regression line must pass through the mean point $(\bar{x}, \bar{y}) = (14.5, 72.0)$: $72.0 = 2.4(14.5) + c \Rightarrow 72.0 = 34.8 + c \Rightarrow c = 72.0 - 34.8 = 37.2$. Thus $y = 2.4x + 37.2$. (b) The slope 2.4 indicates that for each additional hour of study, a student's predicted exam mark increases by 2.4 marks. (c) For $x = 18$: $\hat{y} = 2.4(18) + 37.2 = 43.2 + 37.2 = 80.4$ marks.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.

1. Substitute $\sin^2\theta = 1 - \cos^2\theta$: $2(1 - \cos^2\theta) + 3\cos\theta - 3 = 0 \Rightarrow 2 - 2\cos^2\theta + 3\cos\theta - 3 = 0$. $-2\cos^2\theta + 3\cos\theta - 1 = 0 \Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$. 2. Factor the quadratic in $\cos\theta$: $(2\cos\theta - 1)(\cos\theta - 1) = 0$. 3. Solve each branch: Branch 1: $2\cos\theta - 1 = 0 \Rightarrow \cos\theta = \frac{1}{2}$. Reference angle: $\alpha = \frac{\pi}{3}$. In $[0, 2\pi]$, cosine is positive in Q1 and Q4.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the logarithmic equation $\log_3(x + 5) + \log_3(x - 1) = 2 + \log_3(2)$ for all real values of x.

1. Domain constraints: $x + 5 > 0 \Rightarrow x > -5$ and $x - 1 > 0 \Rightarrow x > 1$. Thus $x > 1$. 2. Rewrite the right-hand side using $2 = \log_3(3^2) = \log_3(9)$: $2 + \log_3(2) = \log_3(9) + \log_3(2) = \log_3(18)$. 3. Combine the left-hand side: $\log_3((x + 5)(x - 1)) = \log_3(18) \Rightarrow (x + 5)(x - 1) = 18$. 4. Expand and solve the quadratic equation: $x^2 + 4x - 5 = 18 \Rightarrow x^2 + 4x - 23 = 0$. 5. Quadratic formula.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 10.01 Series & Binomial Theorem

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{9^2 + 10^2 - 7^2}{2(9)(10)} = \frac{81 + 100 - 49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333$. $A = \arccos(11/15) \approx 42.8^\circ$. Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7 + 9 + 10}{2} = 13$..

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 11.01 Kinematics**

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 12.01 Forces & Equilibrium**

The sum of the first n terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term u_1 and the second term u_2. (b) Find the common difference d. (c) Find an explicit expression for the nth term u_n, and calculate u_{25}.

(a) $u_1 = S_1 = 4(1)^2 - 3(1) = 1$. $S_2 = 4(2)^2 - 3(2) = 16 - 6 = 10$. $u_2 = S_2 - S_1 = 10 - 1 = 9$. (b) Common difference: $d = u_2 - u_1 = 9 - 1 = 8$. (c) General term: $u_n = u_1 + (n-1)d = 1 + (n-1)(8) = 8n - 7$. Alternatively: $u_n = S_n - S_{n-1} = (4n^2 - 3n) - [4(n-1)^2 - 3(n-1)] = 8n - 7$. For $n = 25$: $u_{25} = 8(25) - 7 = 200 - 7 = 193$.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws**

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Representation of Data**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Permutations & Combinations**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J ✓ **b.** 480 J

c. 415.7 J **d.** 120 J

Work done $W = Fd\cos\theta = 60 \times 8 \times \cos(60^\circ) = 480 \times 0.5 = 240$ J.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10$ m/s^2, what is the work done against gravity?

a. 600 J ✓ **b.** 150 J

c. 300 J **d.** 60 J

$W = mgh = 15(10)(4) = 600$ J.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?

a. 175 J ✓ **b.** 150 J

c. 350 J **d.** 70 J

$W = F_f \times d = 35 \times 5 = 175$ J.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J ✓ **b.** 4480 J

c. 560 J **d.** 280 J

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(70)(64) = 35 \times 64 = 2240$ J.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10\ \text{m/s}^2$, find its speed at the bottom of the hill.

a. 20 m/s ✓

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

Conservation of energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2(10)(20)} = \sqrt{400} = 20$ m/s.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N ✓

b. 3000 N

c. 9000 N

d. 12000 N

Work-energy principle: $Fd = \frac{1}{2}mv^2 \Rightarrow F(0.15) = \frac{1}{2}(0.02)(90000) = 900 \Rightarrow F = 900/0.15 = 6000$ N.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10\ \text{m/s}^2$, how much energy was lost due to friction?

a. 6000 J ✓

b. 12000 J

c. 18000 J

d. 3000 J

Initial potential energy $E_p = mgh = 60(10)(30) = 18000$ J. Final kinetic energy $E_k = \frac{1}{2}(60)(400) = 12000$ J. Energy lost $= 18000 - 12000 = 6000$ J.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10$ m/s^2).

a. $\sqrt{80} \approx 8.94$ m/s ✓

b. 10 m/s

c. 8 m/s

d. 12 m/s

Work done by gravity $= mgh = 4(10)(6) = 240$ J. Work against friction $= 8 \times 10 = 80$ J. Net kinetic energy $= 240 - 80 = 160$ J. $\frac{1}{2}(4)v^2 = 160 \Rightarrow 2v^2 = 160 \Rightarrow v^2 = 80 \Rightarrow v = \sqrt{80} \approx 8.94$ m/s.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10$ m/s^2, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m ✓

b. 10.0 m

c. 5.50 m

d. 8.25 m

Deceleration up slope: $a = g(\sin 30^\circ + \mu \cos 30^\circ) = 10(0.5 + 0.2 \times 0.866) = 10(0.5 + 0.1732) = 6.732$ m/s^2. Distance $s = u^2/(2a) = 100/(2 \times 6.732) \approx 7.43$ m.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N ✓

b. 1200 N

c. 1350 N

d. 2000 N

$P = Fv \Rightarrow 45000 = F(30) \Rightarrow F = 1500$ N. Since speed is constant, driving force equals resistance: $R = 1500$ N.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. $0.625\ \text{m/s}^2$ ✓ **b.** $0.75\ \text{m/s}^2$

c. $1.00\ \text{m/s}^2$ **d.** $0.50\ \text{m/s}^2$

Driving force at 20 m/s: $F = P/v = 45000/20 = 2250$ N. Net force $= 2250 - 1500 = 750$ N. Acceleration $a = 750/1200 = 0.625\ \text{m/s}^2$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10\ \text{m/s}^2$, find the resistance to motion.

a. 1500 N ✓ **b.** 2500 N

c. 4000 N **d.** 2000 N

Driving force $F = P/v = 60000/15 = 4000$ N. Gravity component along slope $= mg\sin\alpha = 5000(10)(1/20) = 2500$ N. At steady speed:

$F = mg\sin\alpha + R \Rightarrow 4000 = 2500 + R \Rightarrow R = 1500$ N.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A force of 8 N acts on a body, moving it a distance of 7 m in the direction of the force. Find the work done.

a. **56** ✓ **b.** 63

c. 112 **d.** 49

Applying Energy, Work & Power principles, $W = Fd = 8 \times 7 = 56$ J Therefore, the verified correct option is 56.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A car of mass 1000 kg increases speed from 10 m/s to 20 m/s. The work done is:

a. 50 kJ **b.** 100 kJ

c. 150 kJ ✓ **d.** 200 kJ

$W = \frac{1}{2}(1000)(20^2 - 10^2) = 500 \times 300 = 150000\ \text{J} = 150\ \text{kJ}$

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

An engine develops a power of 12483 W while a vehicle moves at a constant 19 m/s. Find the driving force exerted.

a. **657** ✓

b. 647

c. 1314

d. 667

$P = Fv \Rightarrow F = \frac{P}{v} = \frac{12483}{19} = 657$ N

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the change in gravitational potential energy of a body of mass 18 kg raised through a height of 15 m. Take $g = 9.8$ m/s^2.

a. 5292.00

b. 2664.00

c. **2646.00** ✓

d. 2628.00

$\Delta GPE = mg\Delta h = 18 \times 9.8 \times 15 = 2646.00$ J

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the kinetic energy of a body of mass 16 kg moving at 7 m/s.

a. 376

b. 408

c. **392** ✓

d. 784

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(16)(7)^2 = 392$ J

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 13.01 Newton's Laws

In how many ways can 8 different books be arranged in a row on a shelf?

a. 45808

b. 35108

c. 40625

d. **40320** ✓

Applying Permutations & Combinations principles, $8! = 40320$ Therefore, the verified correct option is 40320.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 15.01 Representation of Data

In how many ways can 6 different people be seated in a row of 6 chairs?

a. **720 ✓**

b. 817

c. 800

d. 725

Applying Permutations & Combinations principles, $6! = 720$ Therefore, the verified correct option is 720.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 16.01 Permutations & Combinations

Find the gradient of the normal to $y = 3x^2 + 2x$ at $x = -1$.

a. -3/4

b. -1/4

c. 5/4

d. **1/4 ✓**

Tangent gradient $= \frac{dy}{dx}\Big|_{x=-1} = -4$. Normal gradient $= -\frac{1}{-4} = 1/4$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable X has distribution $P(X=1) = 1/10, P(X=2) = 2/5, P(X=3) = 1/2$. Find $\text{Var}(X)$.

a. 9/13

b. **11/25 ✓**

c. 26/37

d. 9/10

$E(X) = 12/5, E(X^2) = 31/5$. $\text{Var}(X) = E(X^2) - [E(X)]^2 = 31/5 - (12/5)^2 = 11/25$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 18.01 Binomial Distribution

Given $X \sim N(62, 5^2)$, find the value a such that $P(X < a) = 0.05$, correct to 2 decimal places.

a. 51.28

b. 56.28

c. 70.22

d. **53.78 ✓**

From tables, $z = \Phi^{-1}(0.05) \approx -1.6449$. $a = \mu + z\sigma = 62 + (-1.6449)(5) \approx 53.78$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 19.01 Normal Distribution

A histogram class has width 5 and frequency 21. Find the frequency density.

a. 16/5

b. 42/5

c. 26/5

d. **21/5** ✓

Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{21}{5} = 21/5$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the equation of the circle with centre $(3, -5)$ and radius 2.

a. $(x-3)^2+(y+5)^2=4$ ✓

b. $(x+3)^2+(y-5)^2=4$

c. $(x-3)^2+(y+5)^2=2$

d. $(x-3)^2+(y+5)^2=8$

Circle formula $(x-a)^2+(y-b)^2=r^2$ with $(a,b)=(3,-5)$, $r=2$: $(x-3)^2+(y+5)^2=4$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Graphs & Transformations

Find the x-coordinates of the stationary points of $y=3x^3-9/2x^2-54x$.

a. $x=-2$ or $x=-3$

b. $x=-2$ or $x=3$ ✓

c. $x=-1$ or $x=3$

d. $x=2$ or $x=3$

$\frac{dy}{dx}=9x^2-9x-54=3(3)(x-3)(x+2)=0\Rightarrow x=-2$ or $x=3$

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